

Optimization

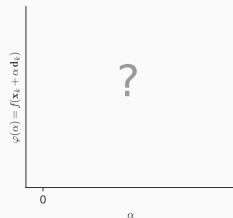
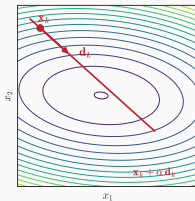
2.1 Univariate optimization: definition, classification and general remarks

José Firmino Aguilar Madeira

Chapter 2. Univariate optimization — why a single variable?

The problems we care about have n variables. Why study optimization in *one*?

Almost all methods of Ch. 3 do, at each iteration, $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$: they choose a **direction** $\mathbf{d}_k$ and then **how far to go**, α_k .



Chapter 2. Univariate optimization — why a single variable?

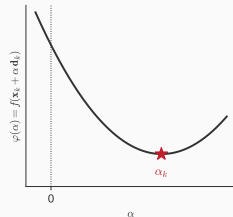
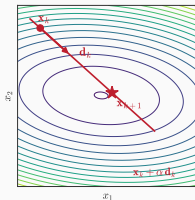
The problems we care about have n variables. Why study optimization in *one*?

Almost all methods of Ch. 3 do, at each iteration, $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$: they choose a **direction** $\mathbf{d}_k$ and then **how far to go**, α_k .

With $\mathbf{x}_k$ and $\mathbf{d}_k$ fixed, $\mathbf{x}_k + \alpha \mathbf{d}_k$ is a **line**, and along it

$$\varphi(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{d}_k)$$

is a function of **one variable**. Choosing α_k means minimizing φ : the **line search**.



Chapter 2. Univariate optimization — why a single variable?

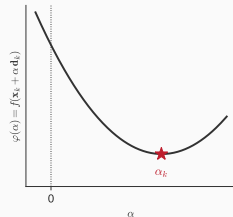
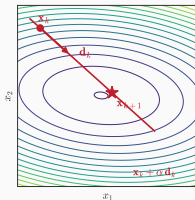
The problems we care about have n variables. Why study optimization in *one*?

Almost all methods of Ch. 3 do, at each iteration, $\mathbf{x}_{k+1} = \mathbf{x}_k + \alpha_k \mathbf{d}_k$: they choose a **direction** $\mathbf{d}_k$ and then **how far to go**, α_k .

With $\mathbf{x}_k$ and $\mathbf{d}_k$ fixed, $\mathbf{x}_k + \alpha \mathbf{d}_k$ is a **line**, and along it

$$\varphi(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{d}_k)$$

is a function of **one variable**. Choosing α_k means minimizing φ : the **line search**.



In 1D we decide **how far to go**. In n D we must first decide **where** to go (Ch. 3).
And the line search is repeated at every iteration: in Ch. 3 it is often most of the cost.

The problem: minimum value and minimizer

A **univariate** optimization problem: $\min_{a \leq x \leq b} f(x)$ (the only constraint is the interval).

There are two possible answers:

$$\text{minimum value: } f^* = \min_{a \leq x \leq b} f(x)$$

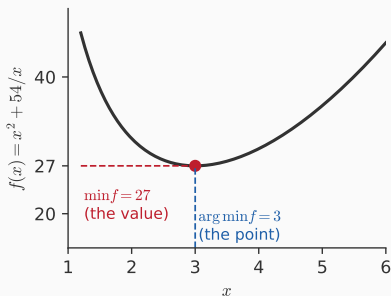
$$\text{minimizer: } x^* = \arg \min_{a \leq x \leq b} f(x)$$

f^* is a value of f ; x^* is the point where it is attained:
 $f(x^*) = f^*$. In the example: $\min f = 27$, $\arg \min f = 3$.

In the line search what matters is the **point**:

$$\alpha_k = \arg \min_{\alpha > 0} \varphi(\alpha).$$

If there are several minimizers, $\arg \min$ denotes the set of them; in this chapter there will be only one.



$f(x) = x^2 + 54/x$: the example that runs through the chapter.

What can we know about f ?

What information do we have? How much does it cost to obtain? What can we guarantee with it?
(the question that runs through the course)

$f(x)$ only	order 0	→ 2.2 golden section: <i>eliminate</i> interval; 2.3 Powell: <i>interpolate</i>
$f(x)$ and $f'(x)$	order 1	→ 2.4 bisection: <i>locate</i> the zero of f'
f , f' and f''	order 2	→ 2.5 Newton: <i>linearize</i> f'
only f , but we want f' , f''		→ 2.6 finite differences: <i>approximate</i> f' , f''

More information: potentially
faster (linear, superlinear, quadratic)

but it **costs**: derivatives are costly,
or unreliable if f is noisy; and it may make
the method **less robust** (no guaranteed
bracketing)

The cost of a method is measured in **evaluations of f** (and of f' , f''), not in arithmetic operations. We count separately n_f (evaluations of f), n_g (of f' ; in $\mathbb{R}^n$, gradients) and n_H (of f'' ; Hessians).

Iterative methods: when to stop?

Except in trivial cases, the minimum is not obtained in a single step. An iterative method:

1. starts from an initial approximation x_0 (or from an interval $[a_0, b_0]$);
2. generates a sequence $x_1, x_2, \dots$ in which each term uses information on f at the previous ones — only at x_k (Newton), or at several: successive quadratic interpolation (Powell's method) uses three;
3. **stops** when a criterion is satisfied.

Usual stopping criteria

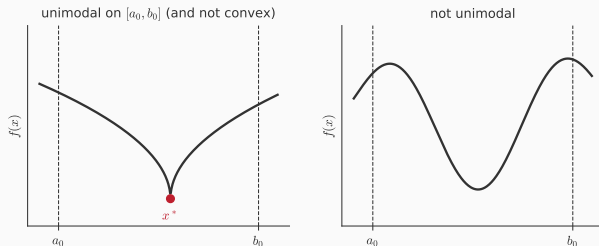
$$\frac{|x_{k+1} - x_k|}{\max(1, |x_k|)} \leq \varepsilon_x, \quad |f(x_{k+1}) - f(x_k)| \leq \varepsilon_f, \quad |f'(x_k)| \leq \varepsilon_g, \quad b_k - a_k \leq \varepsilon_x, \quad k \geq k_{\max}.$$

Always use $k_{\max}$ (it protects against methods that do not converge). Dividing by $\max(1, |x_k|)$ makes the criterion relative when $|x_k|$ is large and absolute near zero; in interval methods the length $b_k - a_k$ is used.

Unimodal functions

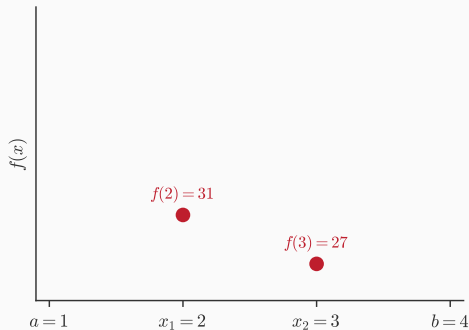
Definition

f is **unimodal** on $[a_0, b_0]$ if it has one and only one local minimum x^* in that interval: it is strictly decreasing on $[a_0, x^*]$ and strictly increasing on $[x^*, b_0]$.



It requires neither continuity, nor derivatives, nor convexity — the “down, then up” shape is enough. Every strictly convex function is unimodal; the converse is false (on the left: unimodal, not convex and not differentiable at x^*).

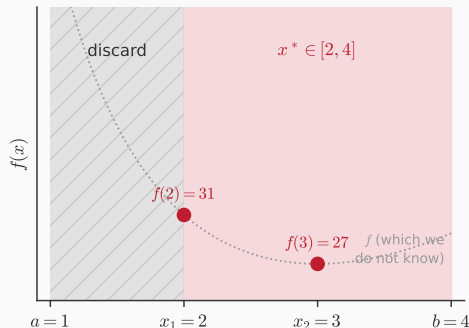
Only two values of f : what can we throw away?



f is unimodal on $[1, 4]$, but **we do not know it**: we only have $f(2) = 31$ and $f(3) = 27$.

Which part of the interval are we sure we do not need?

Only two values of f : what can we throw away?

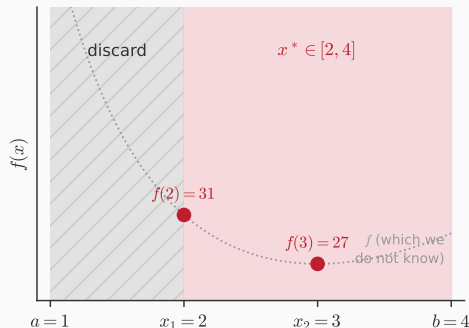


f is unimodal on $[1, 4]$, but **we do not know it**: we only have $f(2) = 31$ and $f(3) = 27$.

Which part of the interval are we sure we do not need?

Since $f(2) > f(3)$, the minimum is not in $[1, 2)$: if it were, f would increase from 2 to 3. Hence $x^* \in [2, 4]$ — one third of the interval discarded with two evaluations and *without derivatives*.

Only two values of f : what can we throw away?



f is unimodal on $[1, 4]$, but **we do not know it**: we only have $f(2) = 31$ and $f(3) = 27$.

Which part of the interval are we sure we do not need?

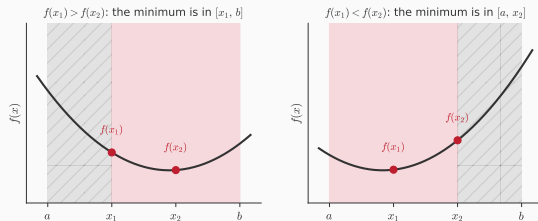
Since $f(2) > f(3)$, the minimum is not in $[1, 2)$: if it were, f would increase from 2 to 3. Hence $x^* \in [2, 4]$ — one third of the interval discarded with two evaluations and *without derivatives*.

And $x_2 = 3$ stays **inside** the new interval, with f already computed.

If we repeat this many times, where should x_1, x_2 be placed to **reuse** an evaluation? $\rightarrow$ 2.2

The general rule: interval elimination

With f unimodal on $[a, b]$ and two interior points $x_1 < x_2$:

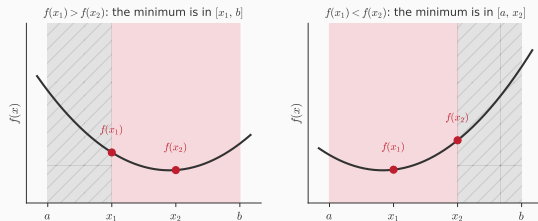


$$f(x_1) > f(x_2) \Rightarrow x^* \in [x_1, b], \quad f(x_1) < f(x_2) \Rightarrow x^* \in [a, x_2], \quad f(x_1) = f(x_2) \Rightarrow x^* \in [x_1, x_2].$$

Robust (guaranteed bracketing): the interval always contains x^* , and shrinks predictably. Only the **order** of the two values is used ($<$ or $>$).

The general rule: interval elimination

With f unimodal on $[a, b]$ and two interior points $x_1 < x_2$:



$$f(x_1) > f(x_2) \Rightarrow x^* \in [x_1, b], \quad f(x_1) < f(x_2) \Rightarrow x^* \in [a, x_2], \quad f(x_1) = f(x_2) \Rightarrow x^* \in [x_1, x_2].$$

Robust (guaranteed bracketing): the interval always contains x^* , and shrinks predictably. Only the **order** of the two values is used ($<$ or $>$).

What if we also used their *magnitude*? With three values of f a parabola can be fitted to estimate where the minimum is $\rightarrow$ 2.3 (Powell).

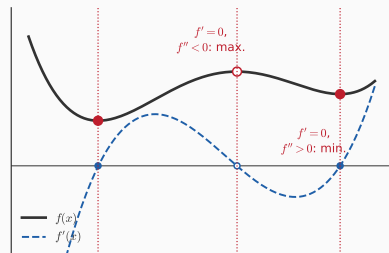
With derivatives: looking for the zero of f'

If f is differentiable, at an interior minimum $f'(x^*) = 0$ (necessary condition); if moreover $f''(x^*) > 0$, x^* is a strict local minimum (sufficient condition).

Instead of minimizing f , we can look for the zero of f' :

- **bisection** on f' (2.4): the *sign* of f' tells on which side the zero is. It keeps an interval with $f'(a) < 0 < f'(b)$ (f' continuous);
- **Newton** on f' (2.5): the slope of f' , that is f'' , lets us *estimate* where the zero is: $x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$.

Caution: $f' = 0$ also at maxima and inflection points — check f'' ($f'' = 0$ is inconclusive).

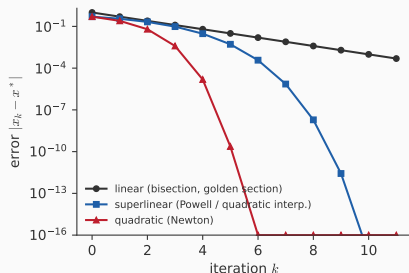


Rate of convergence

Let $e_k = |x_k - x^*|$. The convergence is called

- **linear** if $e_{k+1} \leq c e_k$, $0 < c < 1$ — it gains a fixed number of digits per iteration. In interval methods it is the *length* L_k that decreases linearly ($L_{k+1} = 0.5L_k$ in bisection, $0.618L_k$ in golden section) and $e_k \leq L_k/2$;
- **superlinear** if $e_{k+1}/e_k \rightarrow 0$ (quadratic interpolation: order ≈ 1.3);
- **quadratic** if $e_{k+1} \leq c e_k^2$ — near x^* , the number of correct digits *tends to double* per iteration (Newton).

The price of speed: Newton needs f'' and x_0 close by; under their respective assumptions, bisection and golden section have guaranteed convergence.



Chapter map: five ideas

Method	Information	Idea	Guarantee / speed
2.2 Golden section	f	eliminate interval	robust (f unimodal); linear, 0.618 per evaluation
2.3 Powell	f	interpolate a parabola	fast near the minimum (superlinear); on its own, no guarantee
2.4 Bisection	f'	locate the zero of f'	robust (single bracketed zero); linear, 0.5
2.5 Newton	f', f''	linearize f'	local quadratic; may diverge
2.6 Finite differences	f	approximate f', f''	truncation error + round-off error

In practice they are combined: a robust method (guaranteed bracketing) locates the minimum, a fast one refines it.

golden section + parabolic interpolation → fminbnd
→ line search of Ch. 3 (steepest descent, damped Newton, CG)

Ch. 2 is not just “earlier material”: it explains a component that works *inside* the algorithms of Ch. 3.

KEY TAKEAWAYS — 2.1

1. In $\mathbb{R}^n$, in direction-based methods, one chooses a direction and then how far to go:
 $\alpha_k = \arg \min_{\alpha > 0} \varphi(\alpha)$, with $\varphi(\alpha) = f(\mathbf{x}_k + \alpha \mathbf{d}_k)$ — a problem in **one** variable.
2. $\min f$ is the minimum *value*; $\arg \min f$ is the *point* where it is attained.
3. Methods differ in the **information** they use (f only, f' , f''), in what it costs and in what it guarantees: more information may give faster convergence (linear $\rightarrow$ superlinear $\rightarrow$ quadratic), but it costs and may make the method less robust (no guaranteed bracketing).
4. With f unimodal, **two values** of f are enough to discard part of the interval (only their order is used); with derivatives, minimizing f becomes **looking for the zero of f'** ($f' = 0$ also at maxima: check f'').

Next (2.2): where to place x_1 and x_2 so that each reduction of the interval costs **a single** evaluation of f ?

References Full list: *Course bibliography*, at the end of the course PDF.

- Deb, K., *Optimization for Engineering Design: Algorithms and Examples*, 2nd ed., PHI Learning, 2012 — Ch. 2.
- Nocedal, J., Wright, S. J., *Numerical Optimization*, 2nd ed., Springer, 2006 — Ch. 3, line search.

Resources

- MATLAB: `fminbnd`.